Energy

Track changes

1 Stores → transfers

Energy is conserved. Stores include kinetic, gravitational, elastic, thermal, chemical and nuclear.

Transfers: mechanically, electrically, by heating or radiation. Light and sound are not stores.

2 Calculate store changes

$E_k = \frac{1}{2}mv^2$; $E_p = mg\Delta h$; $E_e = \frac{1}{2}ke^2$. Use kg, m, m/s and N/m.

Spring: extension, not total length; stay within proportional limit. Doubling v or e quadruples energy.

Measure transfers

3 Power & efficiency

60 J useful + 40 J dissipated

$P = E/t = W/t$. Watts = joules per second. Same work in less time → greater power.

Efficiency = useful output / total input; ×100 for %. Dissipated energy is less useful, not destroyed.

4 Heating & insulation

$\Delta E = mc\Delta\theta$. Temperature change, not final temperature. Measure mass, energy and temperature rise.

Insulation: low conductivity and greater thickness reduce transfer. Compare cooling with controls.

Choose resources

5 Resources & reliability

Non-renewable: coal, oil, gas, nuclear. Renewable: biofuel, wind, Sun, tides, waves, hydro, geothermal.

Uses: heating, transport, electricity. Wind/Sun vary; tides are predictable but not continuous.

6 Evaluate a choice

Compare emissions, waste, habitat effects, cost, reliability and suitable sites.

A resource mix helps match demand. Science informs decisions; economic and social choices matter.